

BTC Intraday & Scalping Algorithmic Trading: An Implementation-Ready Research Report

TL;DR

- **No single BTC scalping strategy is reliably profitable on its own once realistic fees and slippage are applied; the edge lives in a *portfolio of regime-gated strategies plus disciplined cost control*.** Build VWAP mean-reversion/pullback, volatility-squeeze breakouts, and EMA-pullback trend-following on 5m/15m first, all filtered by a regime detector and executed primarily with maker (limit/(`POST_ONLY`)) orders.
- **Costs dominate at short timeframes.** With Binance USD[Ⓢ]-M taker fees at ~0.05% (~0.045% with BNB) a 0.3% gross scalp loses ~30% of its edge to a round trip; a strategy must show net-positive expectancy after ~0.09-0.10% all-in round-trip cost (futures) or it is noise.
- **Derivatives "carry/funding" edges are real but decaying and are not scalps:** per the arXiv paper *Cryptocurrency as an Investable Asset Class: Coming of Age* (arxiv.org/html/2510.14435v2), "the annualized Sharpe ratio of the cryptocurrency carry is 6.45 [2020–2025 full sample]. Beginning in 2024, the Sharpe ratio falls to 4.06, and it turns negative in 2025." Use funding/OI/liquidation data as *filters and context*, not standalone alpha. (arXiv)

Key Findings

1. **Short-timeframe edges are small and fragile.** Order-flow imbalance has high information content but a half-life of seconds. Tigro Blanc's study *Meta-Order Flow in Crypto Perps* (Coinmonks/Medium) found BTC "raw alpha is small: BTC around 0.42 bps at 30s... At a 4 bps round-trip taker cost assumption, all tested configurations are negative net" (despite a strong signal: IC $\approx$ 0.10, t-stat up to 6.86). The implication: sub-1m scalping is a latency/maker-rebate game retail cannot win; focus on 5m-15m where moves (0.3-1%+) can clear costs. (Medium) (Medium)
2. **Strategy success is regime-conditional.** Trend/breakout strategies win in trending/expansion regimes and bleed in chop; mean-reversion (VWAP/Bollinger) wins in ranges and gets run over in trends. A regime classifier (ADX + Bollinger-bandwidth + ATR-percentile) that *switches which strategies are live* is the single highest-leverage component.
3. **VWAP is the highest-value single tool for intraday BTC** because institutional execution algorithms transact around it, creating real reaction. It supports both mean-reversion (fade $\pm 2\sigma$ band back to VWAP in ranges) and trend (pullback-to-VWAP entries in trends).
4. **Volatility behaves cyclically** — contraction precedes expansion. Squeeze/breakout systems (Bollinger-bandwidth or ATR-normalized squeeze + channel breakout +

momentum confirmation) are among the most robust families, but suffer false breakouts in chop and require volume confirmation.

5. **Liquidity-sweep / stop-hunt reversals** are a genuine microstructure phenomenon (price spikes past a swing high/low to trigger clustered stops, then reverses). They give tight stops and high R:R but require waiting for *confirmation* (a candle close back inside the level / market-structure shift), not entering on the wick.
6. **Costs and overfitting are the two account-killers.** Backtested max drawdown should be expected to be $\sim 1.5\times$ larger live; backtested Sharpe > 2 typically falls to 1.0–1.5 live; a Sharpe > 3 in backtest is an overfitting red flag. Multiple-testing is brutal: per Bailey, Borwein, López de Prado & Zhu (2014), "with 45 backtests, random chance will produce one Sharpe > 1.5 even if there is no true alpha," and a 2014 study found 44% of published trading strategies failed to replicate on new data. Use walk-forward / combinatorial purged cross-validation and the Deflated Sharpe Ratio. (MLJAR) (Adventuresofgreg)

Details

A. Cost, latency and data reality (the constraints that shape everything)

Binance fees (2026). Per the Binance Futures fee schedule (corroborated by tradersunion.com and datawallet.com), "Binance Futures fees start at 0.02% for makers and 0.05% for takers at VIP 0, with tiered discounts available and an optional 10% reduction when paying with BNB" (the BNB discount applies to USDT-M, ETH/BTC and USDC-M — not COIN-M). So effective futures taker $\approx 0.045\%$, maker $\approx 0.018\%$ with BNB. Spot is 0.10%/0.10% at VIP 0 (25% BNB discount $\rightarrow 0.075\%$). VIP 1 futures requires $\sim \$15\text{M}$ 30-day volume (0.016% maker / 0.04% taker). **Fees are charged on notional, not margin** — a $10\times$ leveraged 10,000 USDT-margin position has 100,000 USDT notional, so a taker round trip at 0.05% ≈ 100 USDT. **Design implication:** prefer (POST_ONLY) maker limit orders; a taker-only strategy must clear $\sim 0.09\text{--}0.10\%$ round trip on futures (more on spot). (Datawallet)

Slippage. On BTC/USDT, market-order slippage averages $\sim 0.01\text{--}0.03\%$ for orders under $\sim \$10\text{K}$ but can reach 0.1–0.5% in volatile moments; for a 0.2–0.3% scalp target, slippage alone consumes 10–30% of gross profit. Model slippage as a *distribution* conditioned on order type, time-of-day, and current volatility/spread — not a flat constant. (Stratbase)

Latency. True HFT order-book scalping ("imbalance velocity" windows $< 300\text{ms}$) is infeasible from retail cloud; every extra $\sim 10\text{ms}$ of latency erodes a meaningful fraction of micro-edge. Co-location near Binance (e.g., AWS Tokyo (ap-northeast-1)) matters only for sub-second strategies — which you should avoid. For 5m/15m systems, a well-hosted in-region cloud VM is sufficient.

Binance API rate limits. Spot REST: 6,000 request-weight/min per IP; futures $\sim 2,400/\text{min}$ on relevant endpoints. Order counts are tracked separately. WebSocket connections accept max 5 incoming messages/sec; the server pings every 3 minutes and disconnects if no pong within 10 min. **Rule:** use WebSocket *streams* for all market data and user-data (fills) — never poll REST for

prices — and reserve REST/WS-API for order placement. Exceeding limits returns HTTP 429, then 418 (IP ban); implement exponential backoff and monitor the `X-MBX-USED-WEIGHT` header.

`Binance + 4`

B. Prioritized strategy families to build first

1. **VWAP mean-reversion + VWAP pullback-trend** (one indicator, two regimes) — highest ROI on build effort.
2. **Volatility squeeze breakout** (Bollinger-bandwidth/ATR + channel + momentum).
3. **EMA pullback trend-following** (9/21 EMA with 50/200 filter + RSI).
4. **Bollinger-band mean reversion** (range regime).
5. **Liquidity-sweep reversal** (microstructure, tight-stop, high-R:R).
6. **Opening-range / session breakout** (London–NY overlap window).
7. **Order-flow imbalance** (as a sizing/execution filter, or maker-only — not standalone taker alpha).
8. **Funding/OI/liquidation context** (filters, not standalone signals).

C. The 25+ concrete strategy specifications

Convention: **TF** = timeframe, **SL** = stop-loss, **TP** = take-profit. Default sizing = fixed-fractional risk (Section E). "Round-trip cost floor" $\approx 0.09\text{--}0.10\%$ futures taker; aim for average win $\geq 2\text{--}3\times$ cost.

1. VWAP Mean-Reversion (Range Fade) — Long & Short TF 5m (signals), 15m context. Data: session VWAP + $\pm 1\sigma/\pm 2\sigma$ bands, ATR(14), ADX(14). Regime: ADX < 20 (ranging). Long: price tags/closes below -2σ AND bullish reversal candle (hammer/engulfing). Short: tags/closes above $+2\sigma$ with bearish candle. Exit: target = VWAP; runner to opposite $\mp 1\sigma$. SL: $0.5\times$ ATR beyond outer band (or beyond wick). TP: scale 50% at VWAP, trail rest. Trailing: BE once halfway to VWAP. Filters: skip if ADX > 25; skip first minute after major news; require spread < $1.5\times$ normal. Freq $\sim 3\text{--}8/\text{day}$. Best: range/low-ADX. Worst: strong trend (price walks the band). Difficulty Low–Med. Backtest: anchor VWAP to UTC daily reset; test 00:00 UTC vs rolling.

2. VWAP Pullback Trend-Following — Long & Short TF 5m/15m. Data: VWAP, EMA(50), ADX. Regime: ADX > 25, price persistently above (long) / below (short) VWAP. Long: buy pullback touching VWAP (to -0.3σ) with rejection candle. Exit: prior swing; or close back through VWAP. SL: $1\times$ ATR beyond VWAP. TP: $1.5\text{--}2.5R$. Trailing: VWAP-following or chandelier ($3\times$ ATR). Filter: trade the London–NY overlap (13:00–16:00 UTC) — filtering a VWAP-pullback to this window historically improved its profit factor from 1.14 to 1.32. Freq 2–5/day. Best: trend. Worst: chop (whipsaw at VWAP). `Stratbase`

3. Bollinger-Band Mean Reversion (RSI-filtered) — Long & Short TF 5m/15m. Data: BB(20,2.0), RSI(14). Long: candle low $\leq$ lower band AND close back above it AND RSI < 25 $\rightarrow$ enter next

open. Short: high $\geq$ upper band AND close below it AND RSI > 75 . Exit: middle band (SMA20); runner to opposite band. SL: beyond signal-candle extreme. TP: middle band (50%), opposite band runner. Trailing: BE after first target. Filter: ranging only (ADX < 20). Freq 3–6/day. Best: sideways. Worst: trend/squeeze breakout (fading a real breakout — the classic failure). (Babypips)

4. Bollinger Squeeze Breakout — Long & Short TF 15m (more reliable than 5m). Data: BB(20,2), Keltner(20,1.5) or BB-bandwidth percentile, ATR, volume. Setup: BB-bandwidth in lowest ~20th percentile of last N bars (squeeze), or BB inside Keltner. Long: close above upper band AND volume $\geq 1.5 \times$ 20-bar avg AND momentum (ROC/MACD) positive. Short: mirror. SL: opposite side of squeeze range (or $1.5 \times$ ATR). TP: measured-move (range height) or 2.5R. Trailing: ATR chandelier ($3 \times$ ATR) past 1R. Filters: volume confirmation mandatory; avoid if ADX > 40 (exhaustion). Freq 1–4/day. Best: contraction $\rightarrow$ expansion. Worst: choppy false-breakout. False signal: low-volume breakout $\rightarrow$ fakeout (use close-confirmation or pullback-entry variant).

5. ATR Channel / Volatility-Compression Breakout — Long & Short TF 15m. Data: ATR(14)/close (normalized vol), Donchian(N), ROC. Setup: normalized ATR below threshold. Long: close breaks N-bar high with positive ROC; Short: breaks N-bar low with negative ROC. SL: $1.5\text{--}2 \times$ ATR (dynamic, not fixed points). TP: $2\text{--}3 \times$ ATR or trail. Filter: momentum agreement. Best: post-compression. Worst: one-sided strong trend $\rightarrow$ over-frequent signals/overtrading. (FMZ)

6. EMA Crossover + RSI Trend (9/21 with 50 filter) — Long & Short TF 15m. Data: EMA(9), EMA(21), EMA(50/200), RSI(14). Long: EMA9 $>$ EMA21 cross AND price $>$ EMA50 AND RSI > 50 . Short: EMA9 $<$ EMA21 AND price $<$ EMA50 AND RSI < 50 . Exit: opposite cross or RSI back through 45/55. SL: below swing / $1.5 \times$ ATR. TP: $1.5\text{--}2$ R or trail. Filter: ADX > 25 to avoid chop whipsaws (dominant failure mode). Freq 1–3/day. Best: trend. Worst: range (MAs intertwine $\rightarrow$ false crosses). Backtest note: don't over-optimize EMA lengths to one period — a 17/43 tuned on a bull run fails when regime changes. (TradingView + 2)

7. Triple-EMA Trend (10/50/200) — Long & Short TF 15m/1h. Long only above EMA200 with EMA10 $>$ EMA50; short only below EMA200 with EMA10 $<$ EMA50. Exit on EMA10/50 re-cross. SL $2 \times$ ATR; TP trail. Best: strong directional days. Worst: chop. (Medium)

8. MACD + VWAP Momentum — Long & Short TF 5m. Data: VWAP, MACD(12,26,9). Long: price above VWAP AND MACD crosses above signal near VWAP support. Short: below VWAP AND bearish MACD cross. Exit: opposite MACD cross or prior S/R. SL behind recent swing. Best: early trend. Worst: chop (MACD whipsaw). (StockGro) (5paisa)

9. Liquidity-Sweep Reversal (Stop-Hunt) — Long & Short TF 5m/15m (1h structure context). Data: swing/equal highs-lows, volume, optional order-block/FVG. Long: price wicks below a clear swing low/equal lows (sweeps sell-side liquidity), then a candle *closes back above* the swept level (acceptance) $\rightarrow$ enter on that close. Short: mirror above a swing high. SL: just beyond the sweep wick (tight). TP: next opposite liquidity pool / prior swing; $2\text{--}4$ R. Filters: significant HTF levels only; require market-structure shift or strong rejection wick; never enter on the wick. Freq 1–

4/day. Best: ranges with clear levels, session opens. Worst: strong trend continuation (the "sweep" was a real breakout) — hence confirmation requirement. Difficulty Med-High (automating level detection is hard). (Alchemy Markets + 2)

10. Opening-Range Breakout (Session) — Long & Short TF 5m. Range = high/low of first 15-30 min from a session anchor (00:00 UTC or 13:00 UTC NY overlap). Long on break above range high with volume; short below. SL = opposite side of range; TP = 1-2× range height; trail rest. Filters: volume surge; skip if range already > 1.5×ATR. Best: volatility-expansion sessions. Worst: inside/low-volume days.

11. Previous-Day High/Low Breakout — Long & Short TF 15m. Long on confirmed close above prior-day high; short below prior-day low. SL beyond level ± ATR; TP next reference. Best: trend days. Worst: range days fading breakouts.

12. RSI(7) Extreme Reversion — Long & Short TF 5m. Fast RSI(7). Long: RSI < 10 then crosses back above 15 (range/with-trend pullback). Short: RSI > 90 crossing below 85. Exit: RSI 50 or VWAP. SL: recent extreme. Filter: range or with-trend only; never counter a strong trend. Best: choppy/range. Worst: trend (RSI pins). Note: default RSI(14)/MACD(12,26,9) are too slow for 1m — use faster lengths on low TFs. (Stratbase)

13. Stochastic Reversal (14,3,3) — Long & Short TF 5m/15m. Long: %K crosses above %D below 20; short: cross down above 80 (with-trend or range). Exit at 50/mid. SL beyond swing. Best: range. Worst: trend.

14. Keltner Channel + RSI Pullback — Long & Short TF 5m. In uptrend (price upper-half of Keltner) buy when RSI rebounds from ~30-40; mirror for shorts. Exit opposite band or RSI 50. SL beyond channel. Best: steady trend. Worst: violent reversals. (StockGro)

15. Volume-Spike Reversal/Continuation — Long & Short TF 1m/5m. Data: volume, VROC (>50% spike). Climactic spike at a level with rejection wick → fade (reversal); spike with breakout close → continuation. SL beyond spike extreme. Filter: combine with VWAP/level context. Best: capitulation/blowoff. Worst: sustained high-volume trends. False signal: mid-trend spikes that continue.

16. Order-Flow Imbalance (maker-only / sizing filter) TF tick/seconds (used to tilt execution). Data: L2 depth top 3-5 levels. Weighted imbalance = (bid-ask depth)/(bid+ask) with weights [1.0, 0.5, 0.25, 0.125, 0.0625], smoothed via EWMA; treat |imbalance| > 0.6-0.65 as directional with a minimum absolute size threshold; cut signal at a fixed 5-30s horizon. **Use:** size larger when book is thick on your side; do NOT use as standalone taker alpha (raw 30s BTC edge ~0.42 bps < ~4 bps round-trip taker cost). Best: maker-rebate market-making overlay or execution timing. Difficulty High. Risks: spoofing, icebergs, latency. (Medium) (Vadim)

17. Book-Resilience Execution Filter Track how fast top-of-book depth recovers after a sweep (EWMA resilience). Resilient book → stops fill near price; fragile (e.g., >30% depth shock for

BTC) → widen stops or stand aside. A gating layer for all strategies, not a signal. Difficulty High.

(Vadim)

18. Funding-Rate Extreme Reversal (context/swing, not scalp) — short bias TF 1h-8h holding.

Data: Binance funding (baseline 0.01%/8h $\approx$ 10.95%/yr; clamp $\pm 0.05\%$ /interval). Rule: when funding $>$ ~95th percentile ($\approx > 0.10$ -0.12%/8h for BTC, "extreme crowding," ~54% annualized cost to hold a long at +0.05%/8h), bias to shorting perpetual / reducing longs, anticipating compression and a 5-15% pullback within 24-72h (practitioner heuristic — Blackperp/Bitget/Sharpe.ai — no rigorous net-of-fee backtest found). Mirror for funding $<$ -0.05% (capitulation → long bias). SL: structure-based, wide. **Caveat:** delta-neutral carry edge has decayed (per arXiv 2510.14435: Sharpe 6.45 full-sample → 4.06 in 2024 → negative in 2025); per Lee et al., *Exploring Risk and Return Profiles of Funding Rate Arbitrage on CEX and DEX* (ScienceDirect S2096720925000818), funding-rate arbitrage Sharpe was -0.21 for Binance BTC and -0.28 for Bitmex BTC (vs positive on smaller markets like Drift XRP at 1.66). Use directionally as a small-sized *filter*. Note historically funding is positive $> 90\%$ of the time (longs pay shorts). (Cube Exchange + 4)

19. Open-Interest / Price Divergence Filter Data: aggregated OI, price, funding. Heuristics: rising price + rising OI + volume = trend continuation; rising price + falling/flat OI = weak move (tighten/skip); falling price + rising OI + rising funding = fragile crowded longs (cascade risk → favor shorts). Gate trend-vs-reversal strategies. **No validated Sharpe** — heuristic only. (Bitget)

20. Liquidation-Cascade Fade (mean-reversion after climax) — long bias on a down-cascade TF 1m-15m. Trigger: large liquidation event producing a vertical move $\sim 2\sigma$ below 20-bar norm. Wait for exhaustion — a volume-spike candle that fails to make a new low (bids absorbing) — then enter for the V-recovery. SL: below the cascade extreme (give room). TP: prior consolidation/mean. **Critical caveat:** "don't catch the falling knife" — most failures come from entering before exhaustion; this is practitioner-validated, not academically backtested. Post-jump microstructure indicators tend to revert within ~ 30 min. For scale: the Oct 10, 2025 cascade wiped $\sim \$19$ B and 1.6M accounts (BTC $\sim \$122$ k $\rightarrow$ $\$105$ k). Difficulty High. Worst: trend-driven continuation crashes. (arxiv + 2)

21. Multi-Timeframe Confirmation Overlay (meta-strategy) Require HTF (1h) trend agreement for any LTF (5m) trend entry; require HTF range for any LTF mean-reversion entry. A *filter wrapper* on specs 1-9 that cuts counter-context trades; reduces frequency ~ 30 -50%, raises win rate.

22. EMA Ribbon Alignment Scalp (5/8/13/21) — Long & Short TF 1m/5m. Long when ribbon stacked ascending ($5 > 8 > 13 > 21$) and price above; short when descending. Exit on ribbon compression/cross. SL beyond ribbon. Best: clean micro-trends. Worst: chop (constant re-stacking).

23. ALMA + Stochastic Scalp — Long & Short TF 1m/5m. ALMA (low-lag MA) for direction + Stochastic for timing. Enter with-trend on stochastic turn from extreme. Exit opposite

extreme/MA.

24. RSI Divergence Reversal — Long & Short TF 15m. Price new low but RSI higher low (bullish divergence) → long; mirror for bearish. Confirm with micro-structure break. SL beyond divergence pivot. **Caveat:** divergence signals have tested poorly on BTC/ETH in some studies — require confirmation; low priority.

25. Anchored-VWAP Confluence — Long & Short TF 5m/15m. Anchor VWAP to a key event (swing high/low, large liquidation bar, session open). Trade pullbacks to the anchored line in trend direction; confluence of multiple AVWAPs (daily+weekly+event) marks high-probability S/R. SL beyond anchor $\pm$ ATR. Best: post-event trends.

26. Grid / Range Oscillation Bot (range regime only) TF 5m within a defined S/R box. Laddered buy limits toward support, sell limits toward resistance; maker-only to earn the spread. Hard kill-switch if price closes outside box. Best: established ranges. Worst: trend breakout (inventory risk) — must have aggregate exposure cap. (Regimesignals)

27. Mean-Reversion Z-Score (statistical) — Long & Short TF 5m/15m. $Z = (\text{price} - \text{SMA}(N)) / \text{rolling } \sigma$. Long when $Z < -2$ (range), exit at $Z \rightarrow 0$; short when $Z > +2$. Time-stop if not reverted within X bars. SL: Z beyond ∓ 3.5 or ATR. Best: range. Worst: trend (Z stays extreme).

(Stoic)

D. Regime detection (the master switch)

Classify every bar as **TRENDING / RANGING / VOLATILE-EXPANSION / COMPRESSION**:

- **ADX(14):** > 25 trending; < 20 ranging. (Monster Trading Systems)
- **Bollinger-bandwidth percentile:** low = compression; expanding = expansion.
- **ATR-percentile** (current vs trailing distribution): high = volatile.
- Optional: return autocorrelation (positive → trending; negative → mean-reverting); a simple voting classifier or HMM across 4-6 signals (require ≥ 4 agree for high confidence).

(Traderssecondbrain) (Regime)

- **Routing:** TRENDING → trend/breakout specs (2,4,5,6,7,8,10,11); RANGING → mean-reversion (1,3,12,13,26,27); VOLATILE → reduce size, widen stops, prefer sweep/liquidation (9,20); COMPRESSION → arm squeeze breakouts (4,5).
- **Caveat:** regime is identified *after* transition completes (lag); transitions are where most losses cluster — combine with multi-timeframe context. (Traderssecondbrain)

E. Risk management & position sizing

- **Per-trade risk:** 0.25–0.5% of equity for scalps (not the 1–2% of swing trading), since you take 5–15 trades/session. At 0.5% risk and 48% win rate, a 10-trade losing streak (which will happen) costs ~5% of capital; at 2% it costs ~18% and likely triggers tilt. (Stratbase) (Stratbase)

- **Daily loss limit:** 1.5-2% → stop for the day. **Weekly circuit breaker:** -5% → pause and review. (Stratbase)
- **Position size formula:** $\text{size} = (\text{equity} \times \text{risk}\%) / (\text{entry} - \text{stop distance})$. Use ATR-based stop distance so size auto-scales with volatility.
- **Kelly:** per-strategy $f^* = (b \cdot p - q) / b$, then use **fractional Kelly** ($\frac{1}{4}$ - $\frac{1}{2}$) — full Kelly on crypto implies 50-80% drawdowns and overestimates edge. Treat Kelly as a *ceiling*, recalc every 50-100 trades per strategy, run separately per strategy (don't blend win rates). (Stratbase + 3)
- **Correlation control:** treat highly correlated open positions (same direction/driver) as one exposure unit; cap aggregate net BTC exposure regardless of how many strategies fire.

F. Strategy scoring, ranking & portfolio management

Per-strategy metrics (rolling, e.g., last 100 trades): expectancy (\$ and R; must exceed costs by 2-3×); profit factor (>1.5 desirable); win rate + avg R:R; Sharpe and Sortino (>1 acceptable, >2 suspicious live); max drawdown, recovery factor (>2), Calmar; **Deflated/Probabilistic Sharpe** to penalize multiple testing; trade frequency & turnover.

Composite score (on live mock/paper, not backtest), e.g.:
$$\text{Score} = 0.35 \cdot \text{norm}(\text{expectancy}) + 0.25 \cdot \text{DSR} + 0.20 \cdot \text{profit_factor} + 0.20 \cdot (1 - \text{norm}(\text{maxDD}))$$
, re-ranked weekly / every N trades so decaying strategies are demoted.

Take-all vs top-ranked: run all *positive-expectancy, regime-appropriate* strategies, but (a) cap capital to the top-K by composite score within each regime, and (b) de-duplicate: if two strategies fire same-direction within a short window on correlated logic, take only the higher-ranked one (or net them). This blends diversification (uncorrelated edges smooth equity) with selection (don't fund decaying strategies).

Allocation: allocate risk budget (not capital) via inverse-volatility / equal-risk-contribution so one high-variance strategy can't dominate; rebalance infrequently to limit cost drag.

G. Software architecture (event-driven, modular)

1. **Market-data ingestor** — WebSocket clients for klines (1m/5m/15m), aggTrades, depth, mark price/funding; normalize to internal events; reconnect/heartbeat (pong within 10 min); REST gap-fill on reconnect.
2. **Feature/indicator engine** — VWAP, EMAs, ATR, ADX, BB, RSI, OFI, regime tags per bar; incremental updates (Welford for z-scores) to avoid recompute.
3. **Strategy workers** — each strategy is an isolated subscriber emitting typed Signal events (confidence, intended size, SL/TP); stateless where possible.
4. **Regime service** — publishes current regime; a router enables/disables workers accordingly.
5. **Portfolio/risk manager** — the single chokepoint: de-dup, correlation/exposure caps, sizing, daily/weekly circuit breakers; approves or rejects orders.

6. **Execution/OMS** — prefer `POST_ONLY` maker; smart re-quote; handle partial fills, 429/418 backoff, idempotent client order IDs; consume user-data stream for fills.
7. **Persistence** — MongoDB (signals, trades, performance, regime snapshots) + Redis for hot state (positions, balances, latest features).
8. **Mock/Paper engine** — mirrors OMS interface, fills against a simulator (Section H).
9. **Scheduler/monitor** — health checks, internal-vs-exchange position reconciliation, alerting.
10. **Dashboard/API** — read models.

Use a message bus (Redis Streams / Kafka / NATS); containerize with Docker; externalize config. Python is pragmatic (pandas/numpy, `ccxt`/native Binance connectors). **Three-mode parity**: the *same* strategy and risk code must run in backtest, paper, and live by swapping only execution/data adapters — guaranteeing what you tested is what you trade.

H. Paper / mock trading module (build before live orders)

- Mirror the live OMS API exactly; consume the *same* live WebSocket feed but route orders to a simulator.
- **Fill model**: for L1/bar data use a probabilistic FillModel (move fill one tick against you with probability p ; model limit non-fill / queue via `prob_fill_on_limit`); for L2 data fill sequentially against actual book depth tracking consumed liquidity. Apply realistic fees (maker vs taker per order type), spread, and a volatility-conditioned slippage *distribution* — not a flat constant. `NautilusTrader` `NautilusTrader`
- Run an **ensemble** of execution assumptions (optimistic/realistic/stress) to bracket outcomes.
- Track **live-vs-backtest drift**: sharp divergence means the backtest was optimistic (stale-side fills, look-ahead) — do not progress to live.

I. MongoDB schema recommendation

Use **time-series collections** (5.0+) for high-frequency measurement data (klines, features, regime snapshots) — they bucket by `metaField`+`timeField`, compress heavily, and auto-index time. Use **standard collections** for entities you update (trades, positions, strategy state).

`OneUptime`

```

// market_data (time-series; metaField:{symbol,tf}; granularity: seconds/minutes)
{ ts, symbol:"BTCUSDT", tf:"5m", o,h,l,c,v, vwap, ema9, ema21, atr, adx, bb_up,
bb_dn, rsi, ofi, regime }

// signals (standard)
{ _id, strategy_id, ts, symbol, tf, side:"long|short", entry_ref, sl, tp,
  confidence, size_intended, regime, features_snapshot:{...},
  status:"emitted|accepted|rejected", reject_reason, mode:"backtest|paper|live" }

// trades (standard; one doc per round trip)
{ _id, signal_id, strategy_id, symbol, side, mode,
  entry_ts, entry_px, entry_qty, entry_order_type:"maker|taker",
  exit_ts, exit_px, exit_reason:"tp|sl|trail|time|manual",
  fees_paid, slippage_entry, slippage_exit, funding_paid,
  gross_pnl, net_pnl, r_multiple, holding_secs, mae, mfe,
  regime_at_entry, leverage, notional }

// strategy_performance (standard; rolling)
{ strategy_id, window:"rolling_100|daily|all", as_of_ts,
  trades_n, win_rate, avg_win_r, avg_loss_r, expectancy_r, expectancy_usd,
  profit_factor, sharpe, sortino, deflated_sharpe, max_dd, recovery_factor,
  turnover, composite_score, rank, enabled, regime_breakdown:{...} }

// regime_snapshots (time-series; metaField: symbol)
{ ts, symbol, regime, adx, bb_bandwidth_pct, atr_pct, autocorr,
  classifier_confidence }

// orders (standard; raw OMS audit trail)
{ _id, client_order_id, signal_id, exchange_order_id, ts_sent, ts_ack, ts_fill,
  type, side, px, qty, filled_qty, status, post_only, retry_count, error_code }

// account_equity (time-series): { ts, equity, balance, open_positions_n, day_pnl,
  week_pnl }

```

Set `expireAfterSeconds` TTL on raw tick/feature collections; downsample older data into summary collections. Index signals/trades on `{strategy_id, ts}` and `{mode, ts}`. `OneUptime`

J. Dashboard views

1. **Portfolio equity curve** (live/paper/backtest overlaid) + current drawdown vs limits.
2. **Per-strategy leaderboard** — composite score, expectancy, PF, Sharpe/DSR, max DD, trades, enabled/disabled, rank trend.
3. **Regime timeline** — current regime + history + active strategies.

4. **Open positions & exposure** — net BTC exposure vs cap, correlation clusters.
5. **Trade blotter** — recent fills with fees/slippage/funding and R-multiple; MAE/MFE distribution.
6. **Cost analytics** — fees + slippage + funding as % of gross PnL per strategy (the "are we beating costs?" view).
7. **Live-vs-backtest drift** monitor per strategy.
8. **System health** — WS connectivity, rate-limit headroom ($(X-MBX-USED-WEIGHT)$), reconciliation status, 429/error counts.
9. **Risk panel** — daily/weekly loss vs circuit breakers, current fractional-Kelly sizing per strategy.

K. Overfitting prevention

- **Walk-forward and Combinatorial Purged Cross-Validation (CPCV)** with purging/embargo — never plain k-fold (it leaks future into past). CPCV shows lower Probability of Backtest Overfitting (PBO) and better Deflated Sharpe than walk-forward alone. (ScienceDirect)
- **Deflated Sharpe Ratio (DSR) and Probabilistic Sharpe (PSR)** to adjust for multiple testing and non-normal returns; per Bailey, Borwein, López de Prado & Zhu (2014), "with 45 backtests, random chance will produce one Sharpe > 1.5 even if there is no true alpha," so track the *number of backtests run* and deflate honestly.
- **Keep free parameters few** relative to trades/data; profits must survive realistic cost assumptions; a new idea should retain most of its in-sample Sharpe out-of-sample. (Aron Groups)
- **Red flags:** Sharpe > 3, profit factor >> 2, "perfect" equity curve, or large IS/OOS gap. Per AQR Capital Management, a moving-average strategy's Sharpe "dropped from 1.2 during backtesting to -0.2 when applied to new data"; a 2014 study found 44% of published strategies failed to replicate. (Adventuresofgreg + 3)
- **Reserve a final untouched hold-out** (~20–30%) for a single end-of-research test; don't iterate on it. (PickMyTrade)
- Prefer an **ex-ante economic rationale** per strategy and **stress/Monte-Carlo scenarios** over a single historical path. (Portfoliooptimizationbook)

Recommendations

Stage 0 — Foundations: Implement the market-data ingestor, feature engine, regime service, MongoDB schema, and the mock/paper engine with a realistic cost+slippage fill model. *Advance when:* the same strategy code runs identically across backtest/paper/live adapters and the cost model reproduces a known historical spread/slippage.

Stage 1 — First three strategies: Ship VWAP mean-reversion (#1), VWAP pullback trend (#2), and Bollinger squeeze breakout (#4), each regime-gated and maker-first. *Advance when:* each shows net-positive expectancy $\geq 2\times$ round-trip cost in paper over ≥ 100 trades and DSR > 0 .

Stage 2 — Portfolio & scoring: Add the risk manager (sizing, circuit breakers, de-dup, exposure caps) and the scoring/ranking service. Add 4–6 more strategies (#3, #5, #6, #9, #10). *Advance when:* portfolio paper Sharpe > 1 net of costs over 4–8 weeks across at least one regime change, with selected strategies' return correlation kept moderate.

Stage 3 — Cautious live (canary): Go live on the top-ranked 2–3 strategies only, at minimum size (0.25% risk/trade), maker-only where possible, with hard daily (-1.5%) and weekly (-5%) kill-switches. *Scale up when:* live tracks paper within tolerance (no large negative drift) for ≥ 4 weeks and max DD stays $< 1.5\times$ backtest. Add strategies/size only as each clears these gates.

Thresholds that change the plan: demote/disable any strategy whose rolling expectancy turns negative or whose live-vs-paper drift exceeds ~ 1 standard error of expectancy; cut total risk if portfolio DD hits $1.5\times$ backtested max; halt new-strategy onboarding if cost analytics show fees+slippage+funding $> \sim 40\%$ of gross PnL.

Caveats

- **No guaranteed profits.** BTC is highly volatile with fast regime changes; every backtested edge degrades, and documented crypto edges (funding carry) have recently turned negative net of fees.
- **Sub-minute scalping is largely a loser's game for retail** after fees, slippage and latency; the order-flow edge has a sub-minute half-life and sits below taker costs (BTC ~ 0.42 bps at 30s vs ~ 4 bps round trip). Favor 5m/15m and maker execution.
- **Derivatives "alpha" claims are weak.** Liquidation-cascade and OI-divergence strategies are practitioner heuristics without rigorous, fee-net, peer-reviewed backtests; vendor "60% cascade probability" claims lack disclosed methodology. Funding carry is a multi-hour/day hold, not a scalp, and its edge is compressing (Sharpe $6.45 \rightarrow 4.06 \rightarrow$ negative; Binance BTC funding-arb Sharpe -0.21 net of fees).
- **Backtests lie by default.** Expect live drawdown $\sim 1.5\times$ backtest and live Sharpe well below backtest; treat any Sharpe > 3 as overfit until proven otherwise.
- **Operational risk is real:** WebSocket disconnects, rate-limit bans (429 $\rightarrow$ 418), partial fills, funding charges and exchange outages can each erase a week of edge. The risk manager and reconciliation layer matter more than any single signal.
- **Leverage amplifies ruin.** Fees on notional plus liquidation risk make high leverage incompatible with survival; size from stop distance, not from available margin.